

A REPORT

on

Industrial Training – I

INTERNSHIP REPORT



Submitted in Partial Fulfillment of the requirements
For the Degree of Bachelor of Technology in CSE

By

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Netaji Subhas Institute of Technology (NSIT), Patna
Bihar Engineering University (BEU), Patna

Academic Session: 2024 – 2028

Training Completed at: C-DAC, Patna
Training Period: 22nd December 2025 – 3rd January 2026

CERTIFICATE OF COMPLETION

This is to certify that the report entitled "Industrial Training — I" is a bonafide record of the work carried out by **Viraj Kumar** (Registration No.: 24105103009), a student of Bachelor of Technology in Computer Science and Engineering (CSE) during the academic session 2024–2028 in the Second Year, Third Semester at Netaji Subhas Institute of Technology (NSIT), Patna.

This report has been submitted in partial fulfillment of the requirements for the award of the Bachelor of Technology (B.Tech) degree under Bihar Engineering University (BEU), Patna.

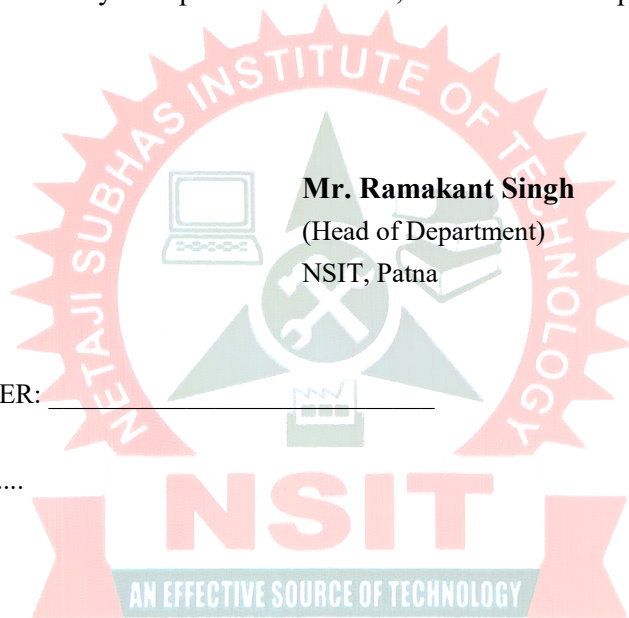
The training was successfully completed at C-DAC, Patna on the topic "**Core Java (OOP Concepts)**".

Ms. J. Dalei
(Academic Dean)
NSIT, Patna

Mr. Ramakant Singh
(Head of Department)
NSIT, Patna

EXTERNAL EXAMINER: _____

Date:



Acknowledgement

I take this opportunity to express my heartfelt gratitude to all those who have supported and guided me throughout my Industrial Training — I, enabling me to successfully complete this report entitled "Industrial Training — I".

Firstly, I would like to extend my sincere thanks to Netaji Subhas Institute of Technology (NSIT), Patna, for providing me with the opportunity to undertake this training, which has been a valuable learning experience in my academic journey.

I would also like to express my sincere gratitude to C-DAC, Patna, for organizing this training program on "Hotel Management System using Core Java" and for providing excellent learning resources and practical exposure.

I am highly thankful to the respected faculty members and trainers, especially Mr. Vaibhav Patle and Ms. Nishu, for their continuous guidance, support, and valuable insights throughout the training period.

I would also like to thank Ms. J. Dalei (Academic Dean) and Mr. Ramakant Singh (Head of Department), NSIT Patna, for their encouragement and support.

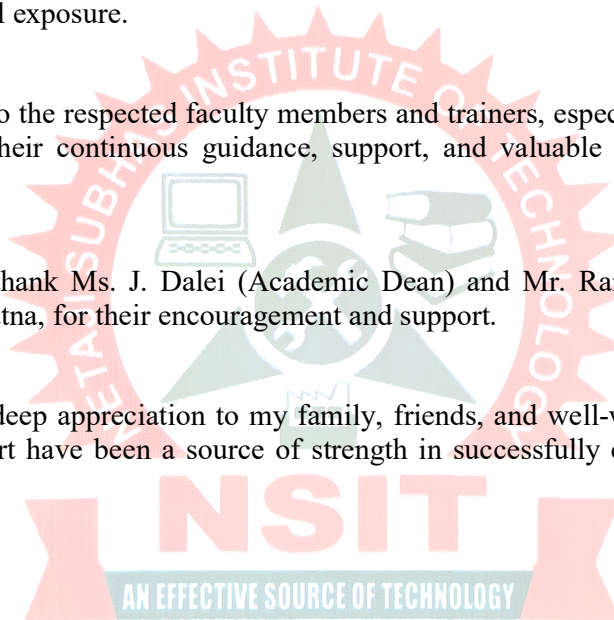
Finally, I extend my deep appreciation to my family, friends, and well-wishers, whose constant motivation and support have been a source of strength in successfully completing this training and report.

Place: Patna

Date:

Student Name: **Viraj Kumar**

Registration No.: 24105103009



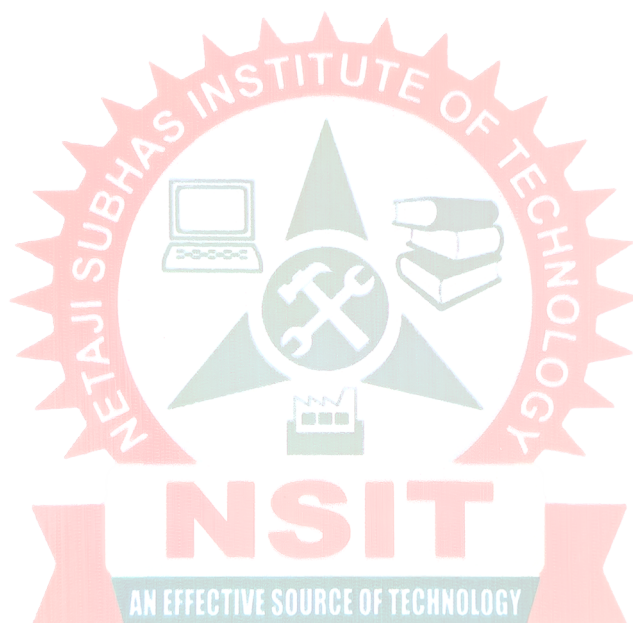
List of Abbreviations

OOP	—	Object-Oriented Programming
JDK	—	Java Development Kit
JVM	—	Java Virtual Machine
JRE	—	Java Runtime Environment
HMS	—	Hotel Management System
SDLC	—	Software Development Life Cycle
GUI	—	Graphical User Interface
CLI	—	Command Line Interface
API	—	Application Programming Interface
IDE	—	Integrated Development Environment
I/O	—	Input/Output
CRUD	—	Create, Read, Update, Delete
CPU	—	Central Processing Unit
RAM	—	Random Access Memory
PRN	—	Participant Registration Number



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CHAPTER 1: INTRODUCTION

1.1 Purpose of the Training

The purpose of this Industrial Training — I was to bridge the gap between theoretical knowledge and practical implementation. In college, we learn many concepts, but this training helped me understand how these concepts are actually used in real-life situations.

This training aimed to provide hands-on experience in programming using Core Java. It helped me improve my coding skills and understand how to write clean and efficient programs. I also learned how different concepts of Object-Oriented Programming (OOP) like classes, objects, inheritance, abstraction, and polymorphism are applied in real software development.

Another important purpose of this training was to develop problem-solving ability and logical thinking. By working on practical tasks and building a real Hotel Management System project, I gained confidence in coding and debugging errors.

Overall, this training helped me understand the basics of software development, improved my technical knowledge, and prepared me for future academic projects and industry work.

1.2 Objectives

The main objectives of the training were:

- To understand the fundamentals of Java Programming Language
- To learn and implement Object-Oriented Programming (OOP) concepts
- To gain practical knowledge of Core Java features such as classes, objects, inheritance, abstraction, and polymorphism
- To develop a real-world Hotel Management System project
- To develop problem-solving and coding skills
- To gain exposure to real-time project development
- To enhance technical and logical thinking abilities

1.3 Overview of Training Program

The Industrial Training — I was conducted at C-DAC, Patna for a duration of two weeks, from 22nd December 2025 to 3rd January 2026. The training sessions were held daily from 9:00 AM to 1:30 PM.

The training focused on "OOP's Concepts using Core Java" and included both theory and practical sessions. In the beginning, basic concepts of Java were taught, such as data types, operators, and control statements. After that, we learned important Object-Oriented Programming (OOP) concepts like classes, objects, inheritance, polymorphism, and encapsulation.

As the training progressed, more advanced topics were covered, including interfaces, exception handling, file handling, collections framework, and multithreading. Each topic was explained

with examples and practical coding, which made it easier to understand and apply in our project — the Hotel Management System.

The sessions were conducted by experienced faculty members, Mr. Vaibhav Patle and Ms. Nishu, who guided us throughout the training. At the end of the training, we worked on the Hotel Management System project and gave a final assessment to test our understanding.

1.4 Company Profile: C-DAC, Patna

The Centre for Development of Advanced Computing (C-DAC) is a well-known research and development organization in India. It works under the Ministry of Electronics and Information Technology (MeitY), Government of India.

C-DAC was established to develop advanced technologies in the field of computing. Over the years, it has played an important role in the growth of Information Technology in India. It is especially known for its work in supercomputing, software development, and technical education.

Location

The training was conducted at C-DAC, Patna, located in Bihar, India. This center provides a good learning environment with proper labs, systems, and experienced faculty members.

Services and Areas of Work

C-DAC works in many advanced areas of technology:

- High-Performance Computing (HPC): Development of supercomputers for research and scientific work
- Software Development: Designing and developing software solutions for various industries
- Cyber Security: Working on data protection, security systems, and ethical hacking
- Artificial Intelligence (AI) & Machine Learning (ML): Developing smart systems and automation technologies
- Embedded Systems & IoT: Working on hardware-based programming and smart devices
- Education and Training: Providing various training programs for students and professionals

Organizational Structure

- Director General: Overall head of C-DAC
- Centre Director: Head of a specific center (e.g., Patna)
- Project Managers / Program Coordinators: Manage projects and training programs
- Faculty / Trainers: Teach students and provide guidance
- Technical Staff: Help in lab work and system management During the training, the sessions were conducted by Mr. Vaibhav Patle and Ms. Nishu, who guided us in both theory and practical work.

CHAPTER 2: TRAINING DETAILS

2.1 Duration of Training

The Industrial Training — I was conducted for a period of two weeks, from 22nd December 2025 to 3rd January 2026 at C-DAC, Patna.

The training sessions were held daily from 09:00 AM to 01:30 PM.

This duration was very helpful to understand both basic and advanced concepts of Java programming in a structured way.

2.2 Department

The training was conducted under the Software Development / Training Department of C-DAC, Patna. This department focuses on providing technical knowledge and practical skills to students in the field of programming and software development. It also helps students to understand how real-world software systems are developed.

2.3 Tools and Technologies Used

During the training, I worked with different tools and technologies related to Core Java:

- Core Java (JDK 8+): Used for writing and running Java programs
- Object-Oriented Programming (OOP): Concepts like abstract classes, interfaces, inheritance, polymorphism, and encapsulation
- Java Collections Framework: ArrayList used to manage room and user data
- Exception Handling: Custom exceptions (HotelException, RoomNotFoundException) to handle runtime errors
- File Handling: FileWriter, BufferedReader, PrintWriter for persistent data storage
- IDE (Eclipse / IntelliJ IDEA): Used to write and execute Java programs

2.4 Training Methodology

The training program was designed in a simple and effective way. Each topic was first explained in theory, and then practical examples were given:

- Trainers explained concepts using real-life examples
- After explanation, coding was done step-by-step
- Students were encouraged to practice on their own
- Doubts were cleared regularly
- A final project (Hotel Management System) was built by applying all concepts

2.5 Daily / Weekly Activities

Week 1 (22nd December – 27th December 2025)

In the first week, the focus was on basic concepts of Java:

- Introduction to Java Programming Language
- Data Types and Operators
- Flow Control Statements (if-else, loops)
- OOP Concepts (Class and Object)

- Constructor and Parameter Passing
- Inheritance in Java
- Abstract Classes and Interfaces
- Packages and Import Statements
- Introduction to Arrays

This week helped me build a strong foundation in Java.

Week 2 (29th December 2025 – 3rd January 2026)

In the second week, more advanced topics were covered:

- Wrapper Classes and String Handling
- Exception Handling (Custom Exceptions)
- File Handling (FileWriter, BufferedReader, PrintWriter)
- Collections Framework (ArrayList)
- Multithreading and Inner Classes
- Project Work: Hotel Management System
- Final Assessment

During this week, I got practical exposure and worked on the Hotel Management System project with my team.

2.6 Overall Experience

The training experience at C-DAC Patna was very good. The sessions were interactive, and the trainers were supportive. I was able to learn new concepts and improve my practical skills. This training helped me become more confident in Java programming and prepared me for future academic projects and real-world applications.



Fig. 1: Group Photo of Training Participants at C-DAC Patna

CHAPTER 3: PROJECT WORK

3.1 Project Title

Hotel Management System

(Console-Based Application using Core Java)

3.2 Introduction to the Project

During the training, we developed a Hotel Management System using Core Java. It is a console-based application, which means it runs on the command prompt and does not use any graphical interface. The system works in a menu-driven way where users can register, log in, and perform various hotel operations easily.

The main purpose of this project is to manage hotel operations such as room booking, check-in, check-out, room service, and billing. Instead of using a database, all data is stored in simple text files (users.txt, room_bookings.txt, hotel_records.txt). This helped us understand file handling in Java and how data can be saved and managed without using complex database systems.

This project helped us understand how real-world applications are developed using Java. We learned how to apply OOP concepts — including abstract classes, inheritance, interfaces, polymorphism, encapsulation — file handling, collections, and exception handling in one integrated project.

3.3 Objectives of the Project

- To apply Core Java and OOP concepts in a real hotel management project
- To implement abstract classes, inheritance, and polymorphism through Room class hierarchy
- To implement interface-based design using HotelOperations interface
- To implement a secure login and registration system with file persistence
- To develop a room booking, check-in, check-out, and billing system
- To understand file handling for persistent data storage
- To improve coding, problem-solving, and teamwork skills

3.4 Team Structure

This project was developed by a team of 4 members, where each member had a specific role and responsibility. The work was divided into different modules so that the project could be completed in an organized and efficient way.

S.No.	Name	PRN	Role / Module
1	Viraj Kumar	STT-25128071923	Main Application (HotelProject.java) + Interface

			Design (HotelOperations.java)
2	Sumeet Kumar	STT-25128071919	Room Model Classes (Room.java) — Abstract class + StandardRoom, Deluxe, Suite, Penthouse
3	Priyanshu Kumar Tiwari	STT-25128071910	Hotel Business Logic (Hotelservice.java) + Custom Exceptions (HotelException.java)
4	Kumar Yashwardhan	STT-25128071908	Login System (LoginServices.java) + File Handling (FileHandler.java)

Working in a team helped us understand how real software projects are developed in parts. Each member focused on their task, and finally all modules were combined to build the complete system.

3.5 Project Structure

The project follows a proper modular structure where different functionalities are divided into separate Java files. This makes the system easy to understand, manage, and maintain.

Directory Structure:

```

PROJECT1/
├── HotelProject.java      [Viraj Kumar]      Main menu-driven application
├── Room.java             [Sumeet Kumar]      Abstract Room + 4 room type
├── classes
│   ├── Hotelservice.java [Priyanshu Tiwari]   Hotel business logic &
│   │   operations
│   ├── LoginServices.java [Kumar Yashwardhan] User registration & login
│   ├── FileHandler.java   [Kumar Yashwardhan] File I/O (billing & booking
│   │   records)
│   ├── HotelOperations.java [Viraj Kumar]      Interface for hotel operations
│   └── HotelException.java [Priyanshu Tiwari]   Custom exceptions
└── Data Files (auto-generated):
    ├── users.txt          Login credentials
    ├── room_bookings.txt  Active booking records
    └── hotel_records.txt   Transaction/billing history
  
```

Class & Method Interaction Flow:

1. User launches HotelProject.main()
2. LoginServices.Register() / LoginServices.login()
3. On successful login → Hotelservice.menu()
4. reserveRoom() → Room.reserve(name, days)
5. FileHandler.saveBookingInfo(roomNo, name, days, type)
6. Optional: Room.addServiceCharge(amount)
7. freeroom() → Room.cancelReservation(actualDays)
8. Room.calculateBill() = (price × days) + serviceCharges

```

9. FileHandler.saveBill(guestName, totalAmount)
10. FileHandler.deleteBookingInfo(roomNumber)
11. showAvailableRooms() → Room.availability()
12. HotelException / RoomNotFoundException on errors

```

3.6 Technologies Used

- Core Java (JDK 8+)
- Object-Oriented Programming: Abstract classes, interfaces, inheritance, polymorphism
- Collections Framework (ArrayList<Room>, ArrayList<User>)
- Exception Handling (Custom HotelException, RoomNotFoundException, InputMismatchException)
- File Handling (FileWriter, PrintWriter, BufferedReader, FileReader)
- IDE: Eclipse / IntelliJ IDEA

3.7 System Features

The system provides various features that help in managing hotel operations in an easy and efficient way:

Feature 1: User Authentication System

- Register new users with username and password validation (min. 6 characters)
- Secure login with a maximum of 3 attempts; account blocked after failed attempts
- User credentials stored persistently in users.txt
- Default admin account pre-loaded (admin / admin334)

Feature 2: Room Management – 4 Room Types

Room Type	Room Numbers	Price / Night	Description
Standard	101 – 105	₹ 2,000	5 rooms – budget-friendly option
Deluxe	201 – 204	₹ 3,500	4 rooms – comfortable mid-range
Suite	301 – 303	₹ 8,000	3 rooms – premium category
Penthouse	401 – 402	₹ 16,000	2 rooms – luxury top-floor

Feature 3: Room Booking (Check-In)

- View all rooms with type, price, and availability status
- Book any available room by entering room number, guest name, and number of days
- Booking information saved automatically to room_bookings.txt
- Optional room service can be added at the time of check-in

Feature 4: Check-Out & Billing

- Guest can check out by entering room number
- Actual days stayed can differ from booked days (early checkout supported)
- Refund calculated automatically for unused days
- Total Bill = (Price per Night × Days Stayed) + Service Charges
- Bill saved to hotel_records.txt with date/time stamp
- Booking entry removed from room_bookings.txt on checkout

Feature 5: Room Service

No.	Service	Charge
1	Breakfast	₹ 500
2	Lunch	₹ 1,000
3	Laundry	₹ 300

Feature 6: File Handling System

All data is stored persistently in text files:

- users.txt → Login credentials for registered users
- room_bookings.txt → Active booking records (room number, guest name, days, room type)
- hotel_records.txt → Transaction records (guest name, total amount, date/time)

3.8 Working of the Project

1. User launches the application — Welcome screen and main menu are displayed
2. User registers a new account or logs in with existing credentials (max 3 attempts)
3. On successful login, the Hotel Menu is displayed with 5 options
4. User can view all rooms or only available rooms
5. User books a room by providing guest name and number of days; optional room service is added
6. Booking data is saved to room_bookings.txt
7. On checkout, actual days are entered, refund is calculated if early, and a detailed bill is printed
8. Final transaction is saved to hotel_records.txt and booking entry is removed

```
OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  Run: HotelProject + - [ ] [ ] ... | X

PS C:\Users\DELL\Desktop\PROJECT1> & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\DELL\AppData\Roaming\Code\User\workspaceStorage\b4680e3bc2ca15a38300b4512969583c\redhat.java\jdt_ws\jdt.ls-java-project\bin' 'PROJECT1.HotelProject'
* WELCOME TO THE HOTEL SYSTEM 2026 *

--- STARTING MENU ---
1. Register New User
2. Login to System
3. Exit Program
Select an option:
1
ENTER NEW USERNAME :
Sumeet@23
ENTER NEW PASSWORD (min 6 letters):
Sumeet23
USER REGISTER SUCCESSFUL
USER SAVED TO FILE

--- STARTING MENU ---
1. Register New User
2. Login to System
3. Exit Program
Select an option:
2
--- LOGIN SYSTEM ---
ENTER USERNAME:
Sumeet@23
ENTER PASSWORD:
Sumeet23
LOGIN SUCCESSFUL!
```

```
OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  Run: HotelProject + - [ ] [ ] ... | X

PS C:\Users\DELL\Desktop\PROJECT1> & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\DELL\AppData\Roaming\Code\User\workspaceStorage\b4680e3bc2ca15a38300b4512969583c\redhat.java\jdt_ws\jdt.ls-java-project\bin' 'PROJECT1.HotelProject'
* WELCOME TO THE HOTEL SYSTEM 2026 *

WELCOME TO HOTEL MENU :
1. SHOWROOMS 2. BOOKROOMS 3. FREEROOMS 4. SHOW ONLY AVAILABLE ROOMS 5. EXIT
CHOICE OPTION (1-5) : 4

Check Room Availability
Room No: 101 | Type: Standard | Price: 2000.0 | Available: true
Room No: 102 | Type: Standard | Price: 2000.0 | Available: true
Room No: 103 | Type: Standard | Price: 2000.0 | Available: true
Room No: 104 | Type: Standard | Price: 2000.0 | Available: true
Room No: 105 | Type: Standard | Price: 2000.0 | Available: true
Room No: 201 | Type: Deluxe | Price: 3500.0 | Available: true
Room No: 202 | Type: Deluxe | Price: 3500.0 | Available: true
Room No: 203 | Type: Deluxe | Price: 3500.0 | Available: true
Room No: 204 | Type: Deluxe | Price: 3500.0 | Available: true
Room No: 301 | Type: Suite | Price: 8000.0 | Available: true
Room No: 302 | Type: Suite | Price: 8000.0 | Available: true
Room No: 303 | Type: Suite | Price: 8000.0 | Available: true
Room No: 401 | Type: Penthouse | Price: 16000.0 | Available: true
Room No: 402 | Type: Penthouse | Price: 16000.0 | Available: true

WELCOME TO HOTEL MENU :
1. SHOWROOMS 2. BOOKROOMS 3. FREEROOMS 4. SHOW ONLY AVAILABLE ROOMS 5. EXIT
CHOICE OPTION (1-5) : 2
```

```
OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  Run: HotelProject + - [ ] [ ] ... | X

PS C:\Users\DELL\Desktop\PROJECT1> & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\DELL\AppData\Roaming\Code\User\workspaceStorage\b4680e3bc2ca15a38300b4512969583c\redhat.java\jdt_ws\jdt.ls-java-project\bin' 'PROJECT1.HotelProject'
* WELCOME TO THE HOTEL SYSTEM 2026 *

CHECKING-IN
ENTER ROOM NUMBER:
101
Enter Guest Name: Viraj raj
Enter Number of Days: 3
Room 101 reserved successfully for Viraj raj for 3 days
Would you like any Room Service now? (y/n):
y
--- SERVICE MENU ---
1. Breakfast - 500
2. Lunch - 1000
3. Laundry - 300
Enter your choice: 2
Service charge added.

WELCOME TO HOTEL MENU :
1. SHOWROOMS 2. BOOKROOMS 3. FREEROOMS 4. SHOW ONLY AVAILABLE ROOMS 5. EXIT
CHOICE OPTION (1-5) : 4

Check Room Availability
Room No: 102 | Type: Standard | Price: 2000.0 | Available: true
Room No: 103 | Type: Standard | Price: 2000.0 | Available: true
Room No: 104 | Type: Standard | Price: 2000.0 | Available: true
Room No: 105 | Type: Standard | Price: 2000.0 | Available: true
Room No: 201 | Type: Deluxe | Price: 3500.0 | Available: true
Room No: 202 | Type: Deluxe | Price: 3500.0 | Available: true
Room No: 203 | Type: Deluxe | Price: 3500.0 | Available: true
Room No: 204 | Type: Deluxe | Price: 3500.0 | Available: true
Room No: 301 | Type: Suite | Price: 8000.0 | Available: true
Room No: 302 | Type: Suite | Price: 8000.0 | Available: true
Room No: 303 | Type: Suite | Price: 8000.0 | Available: true
Room No: 401 | Type: Penthouse | Price: 16000.0 | Available: true
```

CHAPTER 4: LEARNING OUTCOMES

4.1 Skills Gained

Through this project, several important technical and analytical skills were developed. Strong proficiency in Java programming was achieved, along with improved understanding of Object-Oriented Programming (OOP) concepts such as abstraction (abstract Room class), inheritance (StandardRoom, Deluxe, Suite, Penthouse extending Room), polymorphism (overriding getroomname() and display()), and encapsulation (private fields with getters/setters).

Skills in problem-solving and logical thinking were enhanced by designing features like the billing system, early checkout refund calculation, and room availability management. Additionally, experience was gained in debugging and error handling through custom exceptions (HotelException and RoomNotFoundException), which improved the ability to write efficient and error-free code.

4.2 Knowledge Acquired

This project helped in acquiring in-depth knowledge of how a complete software system is designed and implemented. A better understanding of Java Collections Framework (ArrayList) was gained for efficient room and user data management. Knowledge of file handling techniques was developed, enabling persistent storage of booking records, transaction logs, and user credentials.

The project also provided insights into interface-based design (HotelOperations interface) and how it improves code modularity and scalability. Furthermore, it improved understanding of inner classes (User class inside LoginServices) and how to structure a multi-class Java project.

4.3 Practical Exposure

The project provided valuable hands-on experience in developing a real-world application. It offered practical exposure to the software development life cycle (SDLC), including planning, designing, coding, testing, and debugging. Experience was gained in working with real-time data processing, user input validation, and system interaction.

It also helped in understanding how different components of a system interact with each other — for example, how HotelProject calls LoginServices for authentication and then delegates to HotelService for all hotel operations, while FileHandler silently manages all persistence in the background. Overall, this project bridged the gap between theoretical concepts and practical implementation.

CHAPTER 5: CONCLUSION

5.1 Overall Experience

This project has been a highly enriching and insightful learning experience that contributed significantly to both technical growth and overall development. It provided an opportunity to practically implement theoretical concepts learned during the course, especially in Java programming and Object-Oriented Programming (OOP). By working on this Hotel Management System, a deeper understanding of concepts such as abstraction, encapsulation, inheritance, polymorphism, file handling, collections, and exception handling was achieved.

The entire process of developing the project — from requirement analysis, designing the class hierarchy, coding, testing, to debugging — helped in understanding how real-world applications are built. It improved the ability to break complex problems into smaller modules and solve them step by step. Working on features like room booking, billing, check-out with refund, and user authentication gave practical exposure to real-time application scenarios.

During the project, several challenges were faced, such as handling errors during file operations, managing room state correctly between booking and checkout, and ensuring proper program flow with menu navigation. Overcoming these challenges helped in improving problem-solving skills, logical thinking, and debugging capabilities.

Additionally, this project became easier and more understandable for me because I had already studied Java during my diploma and had good hands-on practice in it. Due to my previous experience and consistent practice, I was able to understand and implement concepts more effectively. In my diploma, I also achieved a 2nd rank in class with an A+ grade, which further reflects my strong foundation and interest in this subject.

Overall, this experience has been extremely valuable in bridging the gap between academic knowledge and practical implementation, making it a strong foundation for future software development work.

5.2 Importance of Training

Industrial training is an essential part of an engineering curriculum as it provides real-world exposure and practical knowledge that cannot be fully gained through classroom learning alone. It plays a vital role in developing technical expertise and understanding how concepts are applied in real industry environments.

Through training, students gain hands-on experience in working with technologies, tools, and frameworks used in the industry. It helps in understanding the Software Development Life Cycle (SDLC), including planning, designing, coding, testing, and deployment. This practical exposure enhances the ability to work on real-time projects and prepares students for professional challenges.

Overall, industrial training is crucial for shaping a student's career. It provides a platform to apply theoretical knowledge, gain practical experience, and develop the skills required to become a competent software developer. This experience not only enhances technical abilities but also builds confidence and prepares for future career opportunities in the field of technology.

CHAPTER 6: REFERENCES

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<https://docs.oracle.com/javase/>
- GeeksforGeeks: <https://www.geeksforgeeks.org>
- W3Schools Java Tutorial:
<https://www.w3schools.com/java/>
- TutorialsPoint Java Guide:
<https://www.tutorialspoint.com/java/>
- Stack Overflow (for problem-solving and debugging support): <https://stackoverflow.com/>
- Class notes and lecture materials provided during academic sessions at C-DAC, Patna
- Practical lab manuals and assignments
- Online video lectures and tutorials on Java programming
- Previous knowledge and hands-on practice gained during diploma studies

